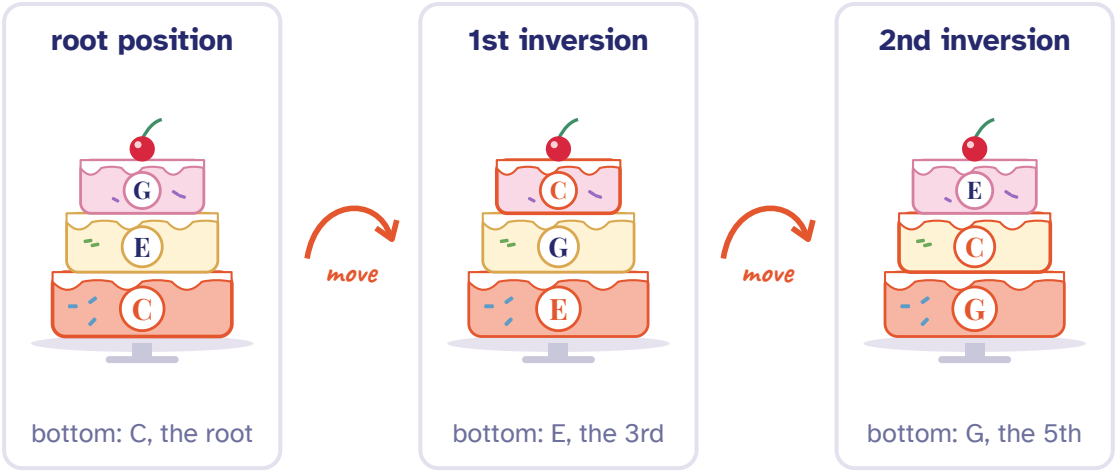
PART IV · CHORDS · 4

Inversions
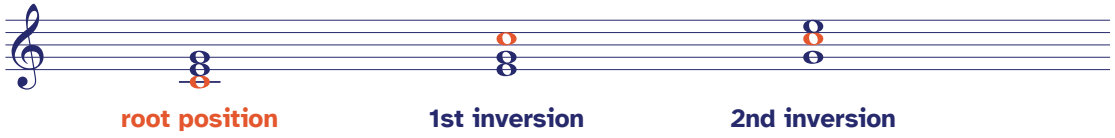
Same three notes, a different one at the bottom. The chord keeps its name; only the bass changes. The red layer is the root, C: watch it travel.

THE SECRET

Move the bottom note to the top: same chord, new bass.



1. Start with the cake **C E G**: root at the bottom.
2. Move the bottom layer to the top: **E G C**. That is the 1st inversion.
3. Move it again: **G C E**, the 2nd inversion. Still **C major**.



The same three chords on the staff; the red note is the root, C. To find the root, restack the notes every other letter (like C E G): then the bottom layer is the root.

Why inversions?

They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

GAME

Which position?

1	2	3
C E G	E G C	G C E

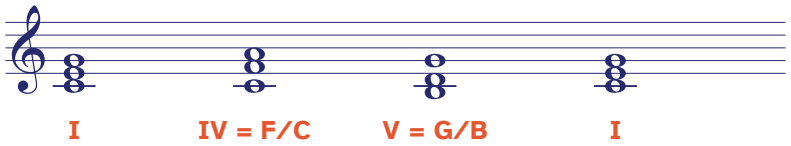
Same three notes. Look at the bottom one: root, 3rd or 5th?

Answer: 1 root position · 2 first inversion (E at the bottom) · 3 second inversion (G at the bottom).

PART IV · CHORDS · 4 · CONTINUED

EXAMPLE 1

Move as little as possible: C, F, G, C



1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note. C steps down to B, F down to D, A down to G: B D G.
3. **I again:** B up to C, D up to E, G stays: C E G.

F stands on C, G stands on B: **inversions** keep the hand in one place.

TRY IT · 10 SECONDS



Climb the chord.

Play C E G, then E G C, then G C E, then C E G an octave up. Same cake, changing its bottom layer as it climbs.

EXAMPLE 2

Write F major in first inversion



1. **Build F major** in root position first.
2. **First inversion:** which layer goes to the bottom, the 3rd or the 5th?
3. **Stack the other two** above it, in order.

The 3rd, A, goes to the bottom: **A C F**.

EXAMPLE 3

Find the second inversion of G



1. **Build G major:** skip one, take the next, from G.
2. **Second inversion:** the 5th goes to the bottom.
3. **Stack the other two** above it.

The 5th, D, at the bottom: **D G B**.

PART IV · CHORDS · 4 · CONTINUED

EXAMPLE 4

Find the root of E G C



1. **Restack the notes** so that each skips one letter: try starting on each note.
2. **Which order works?** C E G, E G C or G C E?
3. **The bottom layer** of that stack is the root.

C E G works: the root is **C**. E G C is C major, 1st inversion.

EXAMPLE 5

Which position? A D F
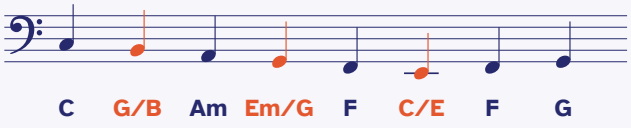


1. **Restack it** so each note skips one letter.
2. **Find the root** (the bottom layer of the stack) and the kind.
3. **Which layer is at the bottom** of A D F: root, 3rd or 5th?

Restacked: D F A, **D minor**. A, the 5th, is at the bottom: **2nd inversion**.

YOU KNOW THIS

Pachelbel's Canon, the bass (here in C)



In the popular version the bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G stands on B, Em on G, C on E.

Try it. Play the bass slowly and hum the famous violin tune over it.

TRAP

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.

SEE ALSO

Chords and triads p. 86 · Chords in a key p. 92 · Slash chords p. 107

PRACTISE IT

Workbook Harmony 4 · Online: Library › Chords